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AI and Fluid Mechanics
31st August – 3rd September 2026

Reinforcement Twinning for Control of Transport-Dominated Systems with Partial Observationsvon

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3. **Twin-Calibrated Policy Search and Replay-Driven Identification**

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   - Model learning with exploration
   - Model exploitation
   - Reinforcement twinning
     (and three ways an imperfect model helps and can be helped)

6. **Conclusions and Outlook**

Active digital twins for actuated systems

An active digital twin **learns from the physical system** while simultaneously **influencing its operation**.

a “steering representation” (Korenhof et al., 2021) or an “acting model” (Wagg et al., 2025)

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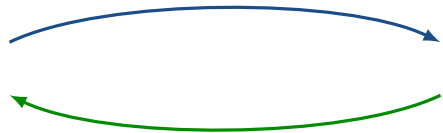
Physical system

Data
trajectories, observations, returns



Actions
control and exploration

Adaptive twin

Control therefore determines both performance and what the twin can learn.

Section 2

General Reinforcement Twinning

Real and Digital Environment: two learning problems

Physical environment •

$$\boldsymbol{s}_{\bullet,k+1} = F_{\bullet}(\boldsymbol{s}_{\bullet,k}, \boldsymbol{z}_k, \boldsymbol{a}_{\bullet,k})$$

$F_{\bullet}$ unknown; $\boldsymbol{\theta}_{\pi}$ **to optimize**

Adaptive twin ○

$$\boldsymbol{s}_{\circ,k+1} = F_{\circ}(\boldsymbol{s}_{\circ,k}, \boldsymbol{z}_k, \boldsymbol{a}_{\circ,k}, \boldsymbol{p}_k)$$

$F_{\circ}$ partially known; $\boldsymbol{\theta}_{p}$ **to identify**

$\boldsymbol{s}$ state · $\boldsymbol{z}$ forcing · $\boldsymbol{a}$ action · $\boldsymbol{p}$ closures · $\boldsymbol{y}$ noisy measurement · $\boldsymbol{o}$ predicted observation

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$$\boldsymbol{s}_{\bullet,k+1} = F_{\bullet}(\boldsymbol{s}_{\bullet,k}, \boldsymbol{z}_k, \boldsymbol{a}_{\bullet,k})$$

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state synchronization: measurements → analyzed twin state

s state · z forcing · a action · p closures · y noisy measurement · o predicted observation

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Control: learn by intervening

$$\boldsymbol{\theta}_\pi^\star = \arg\max_{\boldsymbol{\theta}_\pi} \mathcal{R}_\bullet(\boldsymbol{\theta}_\pi\,;\,\boldsymbol{\theta}_p)$$

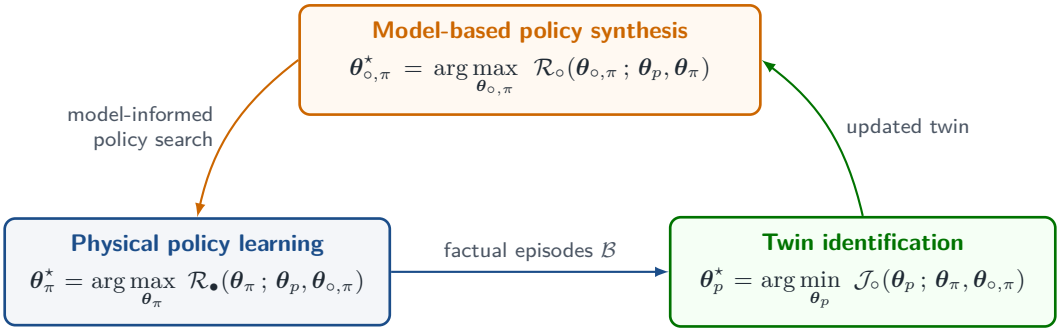
Twin: learn by replaying

$$\boldsymbol{\theta}_p^\star = \arg\min_{\boldsymbol{\theta}_p} \mathcal{J}_\circ(\boldsymbol{\theta}_p\,;\,\boldsymbol{\theta}_\pi)$$

$\boldsymbol{s}$ state · $\boldsymbol{z}$ forcing · $\boldsymbol{a}$ action · $\boldsymbol{p}$ closures · $\boldsymbol{y}$ noisy measurement · $\boldsymbol{o}$ predicted observation

Reinforcement Twinning

Reinforcement Twinning couples physical policy learning and twin identification reciprocally: **three unknowns, each optimised while the other two are held fixed**, and each feeding the other two.



Reinforcement Twinning: algorithm at a glance

Initialize $\mathcal{B}$, $\mathcal{D}_\pi$, $\widehat{\mathcal{I}}^{\mathrm{a}}$, $\hat{\boldsymbol{\theta}}_p$ and the virtual guidance.

1. Select the next interaction e_n

$e_n = 0$: a **performance policy** $\pi_\bullet$ | $e_n = 1$: an **informative excitation** π_ι

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2. Deploy and record

Execute $\pi_\bullet^{(n+1)}$ and store observations, forcing, actions and return as the factual episode $\mathcal{E}^{(n+1)} \to \mathcal{B}$.

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3. Learn from the new experience

Synchronize the twin state; for a performance episode, append $(\boldsymbol{\theta}_\pi, \mathcal{R}_\bullet)$ to $\mathcal{D}_\pi$ and update $\widehat{\mathcal{R}}_\bullet$.

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4. Refresh model-dependent quantities

$\boldsymbol{u}_n$

$u_{p,n}$: re-identify $\hat{\boldsymbol{\theta}}_p$ | $u_{\circ,\pi,n}$: re-synthesize $\bar{\boldsymbol{\theta}}_{\circ,\pi}$ | $u_{R,n}$: recalibrate $\widehat{\mathcal{R}}_\bullet$.

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Section 3

Twin-Calibrated Policy Search and Replay-Driven Identification

Step 1: select the next interaction

1. Select the next interaction

performance $(e_n = 0)$ or information $(e_n = 1)$

2. Deploy and record

the only step that touches the physical system

3. Learn from the new experience

file the episode; synchronize; refit the return surrogate

4. Refresh model-dependent quantities

identify | re-synthesize | recalibrate

Most of this section lives here: what makes a candidate worth trying, and what makes an experiment informative.

Selecting the next interaction: two purposes, two criteria

At physical iteration n, the schedule e_n fixes the **purpose** of the next episode

$e_n = 0$ **performance**

$$\max_{\boldsymbol{\theta}_\pi \in \mathcal{Q}^{(n)}} \mathcal{A}_{\mathrm{RT}}^{(n,m)}(\boldsymbol{\theta}_\pi)$$

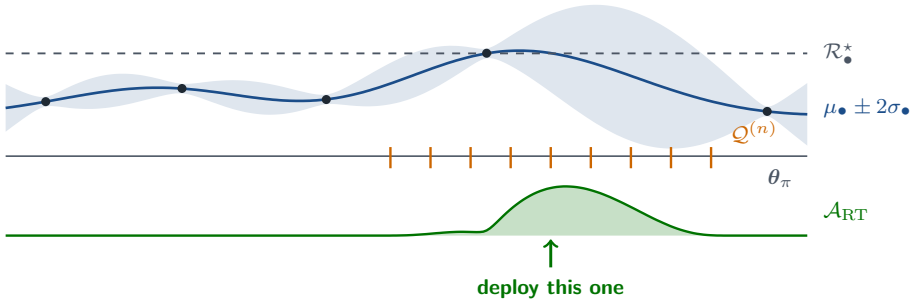
which policy is most likely to perform best on the plant?

First: the performance branch. Both are decided **entirely on the twin** — no physical episode is spent choosing.

There, the action follows the **scope of the run**: recorded actions **replayed**, or the loop **closed on the twin's own observation** — $\pi_\bullet$ to evaluate a candidate, $\pi_\circ$ to synthesize one.

Ranking candidates: expected improvement

$$\mathcal{A}_{\mathrm{RT}}(\boldsymbol{\theta}_\pi) = \mathbb{E}\Big[\max\left(0,\ \widehat{\mathcal{R}}_\bullet(\boldsymbol{\theta}_\pi) - \mathcal{R}^\star_\bullet - \xi\right)\Big] = \Delta_\bullet\,\Phi(\eta_\bullet) + \sigma_\bullet\,\phi(\eta_\bullet), \qquad \boldsymbol{\theta}_\pi \in \mathcal{Q}^{(n)}$$



- how much a candidate could **beat the best so far**
- wide uncertainty beats a merely good mean
- maximized over the finite pool $\mathcal{Q}^{(n)}$

What the twin contributes: a trend, and a region

1. a prior mean for $\widehat{\mathcal{R}}_\bullet$, hence for $\mu_\bullet$ and $\sigma_\bullet$

$$\mathcal{R}_\bullet(\boldsymbol{\theta}_\pi) = \underbrace{\gamma_{\circ,0} + \gamma_{\circ,1}\,\mathcal{R}_\circ(\boldsymbol{\theta}_\pi)}_{\text{calibrated twin trend}} + \underbrace{\delta_\circ(\boldsymbol{\theta}_\pi)}_{\text{discrepancy } \mathcal{GP}} + \varepsilon_\bullet$$

$\gamma_{\circ,0}$ removes bias and $\gamma_{\circ,1}$ sets the scale — both **fitted from the measured physical returns**, so the weight given to the twin is inferred, not declared.

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2. a region in which to propose candidates

$$\mathcal{Q}^{(n)} \subset \left\{ \boldsymbol{\theta}_\pi \in \Theta_\pi : \left\| \widetilde{\boldsymbol{\theta}}_\pi - \widetilde{\boldsymbol{\theta}}^{(m)}_{\circ,\pi} \right\|_2 \leq \varrho^{(m)} \right\}$$

a ball around the **twin-synthesized reference policy** $\boldsymbol{\theta}^{(m)}_{\circ,\pi}$, measured in *whitened* coordinates $\widetilde{\boldsymbol{\theta}}_\pi$: each coordinate is normalized and then divided by an ARD length scale fitted on the twin, so that directions the twin says matter little contribute less to the distance.

Selecting the next interaction: the other branch

At physical iteration n, the schedule e_n fixes the **purpose** of the next episode

$e_n = 0$ **performance**

$$\max_{\theta_\pi \in \mathcal{Q}^{(n)}} \mathcal{A}_{\mathrm{RT}}^{(n,m)}(\theta_\pi)$$

which policy is most likely to perform best on the plant?

$e_n = 1$ **information**

$$\max_{\boldsymbol{\theta}_\iota \in \Theta_\iota} \lambda_{\min}\left[\overline{\boldsymbol{F}}^{(n,m)}(\boldsymbol{\theta}_\iota)\right]$$

which excitation would best reveal the latent $\boldsymbol{\theta}_p$?

Now the information branch: a high-performing policy can be completely uninformative, if it stays where different parameter values produce indistinguishable observations.

Excite what you cannot see

How much would an episode actually *tell* us about $\boldsymbol{\theta}_p$? Score an excitation by how strongly the observations it would produce respond to the latent parameters — the **ensemble Fisher information matrix**.

$$\overline{\boldsymbol{F}}(\boldsymbol{\theta}_\iota) = \frac{1}{N_d} \sum_{\ell=1}^{N_d} \boldsymbol{S}_\ell^\top \, \boldsymbol{\Sigma}_{o,N}^{-1} \, \boldsymbol{S}_\ell, \qquad \boldsymbol{S}_\ell = \frac{\partial \, \widehat{\boldsymbol{o}}_{\circ,0:N}^{(\ell)}}{\partial \, \widetilde{\boldsymbol{\theta}}_p}$$

$\boldsymbol{\Sigma}_{o,N} = \boldsymbol{I}_{N+1} \otimes \boldsymbol{\Sigma}_o$: covariance of the *stacked* observation errors over the episode

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Design: an E-optimal criterion

- maximizing $\lambda_{\min}(\overline{\boldsymbol{F}})$ excites the **least observable** direction
- a trace or determinant criterion could instead be raised by directions that are already well informed

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And which coordinates are identifiable?

$$\overline{F}_{i|\sim i} = \overline{F}_{ii} - \overline{\boldsymbol{F}}_{i,\sim i} \overline{\boldsymbol{F}}^{+}_{\sim i,\sim i} \overline{\boldsymbol{F}}_{\sim i,i}$$

the **marginal Fisher information** of coordinate i: it **measures identifiability**. Zero means *not identifiable*.

Step 3: learn from the new experience

1. Select the next interaction

performance $(e_n = 0)$ or information $(e_n = 1)$

2. Deploy and record

the only step that touches the physical system

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file the episode; synchronize; refit the return surrogate

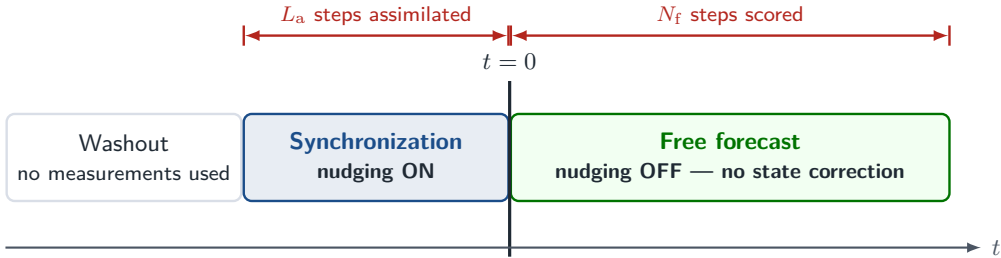
4. Refresh model-dependent quantities

identify | re-synthesize | recalibrate

Every episode, without touching the twin:

$\mathcal{B} \leftarrow \mathcal{B} \cup \mathcal{E}^{(n+1)}, \quad \widehat{\boldsymbol{s}}^{\,\mathrm{a}} \leftarrow \Psi_{\mathrm{o}}(\mathcal{W}_n^-), \quad$ if $e_n = 0$: $\mathcal{D}_\pi \leftarrow \mathcal{D}_\pi \cup (\boldsymbol{\theta}_\pi, \mathcal{R}_\bullet)$ and refit $\widehat{\mathcal{R}}_\bullet$

Synchronize first — then predict freely



nudging, for $t < 0$ only

$$\boldsymbol{\nu}_{k+1} = \boldsymbol{y}_{k+1} - h\left(\boldsymbol{s}^{\mathrm{f}}_{\mathrm{o},k+1}\right), \qquad \boldsymbol{s}^{\mathrm{a}}_{\mathrm{o},k+1} = \mathcal{P}_{\mathrm{adm}}\left(\boldsymbol{s}^{\mathrm{f}}_{\mathrm{o},k+1} + \boldsymbol{K}_{k+1}\boldsymbol{\nu}_{k+1}\right)$$

One episode, one budget: steps spent synchronizing cannot be scored.

Step 4: refresh what depends on the twin

1. Select the next interaction

performance ($e_n = 0$) or information ($e_n = 1$)

2. Deploy and record

the only step that touches the physical system

3. Learn from the new experience

file the episode; synchronize; refit the return surrogate

4. Refresh model-dependent quantities

identify | re-synthesize | recalibrate

Three independent schedules $\boldsymbol{u}_n$ — **identify**, **re-synthesize**, **recalibrate** — decoupling expensive model work from the physical-interaction rate.

Replay-driven identification

The twin re-lives the stored episodes: same forcing, same actions.

$$\mathcal{J}_{\circ}(\boldsymbol{\theta}_p;\mathcal{B}_{\mathrm{tr}}) = \mathcal{J}^{\mathrm{f}}_{\circ,\mathrm{dyn}} \;+\; c_{\mathrm{rew}}\,\mathcal{J}^{\mathrm{f}}_{\circ,\mathrm{rew}}$$

trajectory error

$$\mathcal{J}^{\mathrm{f}}_{\circ,\mathrm{dyn}} = \mathbb{E}_{\mathcal{B}}\left[\frac{1}{N_{\mathrm{f}} n_{\mathrm{id}}}\sum_{k>k_0} \boldsymbol{\nu}_k^{\mathrm{f}\top}\boldsymbol{\Sigma}_{\mathrm{id}}^{-1}\boldsymbol{\nu}_k^{\mathrm{f}}\right]$$

$\boldsymbol{\nu}_k^{\mathrm{f}} = \boldsymbol{y}_k - h(\boldsymbol{s}^{\mathrm{f}}_{\circ,k})$, on the *free* run

return mismatch

$$\mathcal{J}^{\mathrm{f}}_{\circ,\mathrm{rew}} = \mathbb{E}_{\mathcal{B}}\left[\left(\frac{\mathcal{R}^{\mathrm{f}}_{\circ}(\boldsymbol{\theta}_p) - \mathcal{R}_{\bullet}}{s_R}\right)^2\right]$$

what the twin would have *earned*

Every replay is **policy-consistent** (the actions actually applied), **state-consistent** (an observation-conditioned initial state) and **causally consistent** (each measurement used only along the action history that produced it).

When an update fires, what is recomputed

identify the twin — BO proposes, AMSGrad refines through the twin
$$\hat{\boldsymbol{\theta}}_p^{(m+1)} = \arg\min_{\boldsymbol{\theta}} \; \mathcal{J}_\circ(\boldsymbol{\theta}; \mathcal{B}_{\mathrm{val}})$$
if $u_{p,n} = 1$

re-synthesize the reference — *same* BO + AMSGrad, warm-started
$$\boldsymbol{\theta}_{\circ,\pi}^{(m)} = \arg\max_{\boldsymbol{\theta}} \; \mathcal{R}_\circ^{(m)}(\boldsymbol{\theta})$$
if $u_{\circ,\pi,n} = 1$

recalibrate the return trend — generalized least squares, no new episode
$$\widehat{\boldsymbol{\gamma}}_\circ = \left(\boldsymbol{H}_\circ^\top \boldsymbol{V}^{-1} \boldsymbol{H}_\circ\right)^{-1} \boldsymbol{H}_\circ^\top \boldsymbol{V}^{-1} \boldsymbol{r}$$
if $u_{R,n} = 1$

design the next excitation — screened over a finite library on the twin
$$\boldsymbol{\theta}_\iota^{(n+1)} = \arg\max_{\boldsymbol{\theta}_\iota} \; \lambda_{\min}\left[\overline{\boldsymbol{F}}_{\mathrm{rob}}(\boldsymbol{\theta}_\iota)\right]$$
if $e_n = 1$

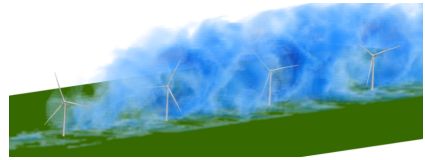
Two searches on the twin, one closed form, and no physical episode

Section 4

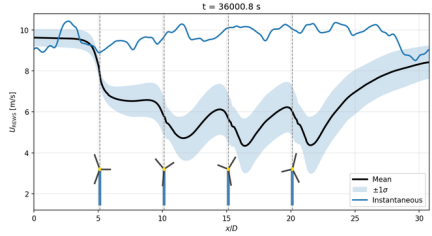
Proposed Test Case: Energy Harvesting from Burgers Flow

A minimal but representative benchmark

- 1D Burgers flow with **localized energy extraction**
- upstream control reshapes the **resource available downstream**
- retains **nonlinearity, delays, partial observability** and device interaction



Dynamic-induction in a wind farm (Schena *et al* 2027)



LES of interacting wakes (Schena *et al* 2027)

A minimal but representative benchmark

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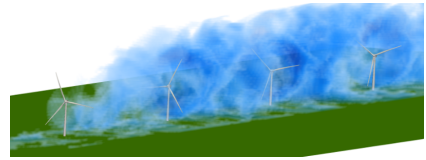
inlet → three extraction devices → outlet



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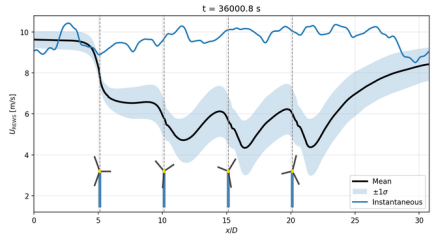

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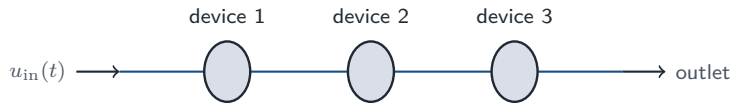
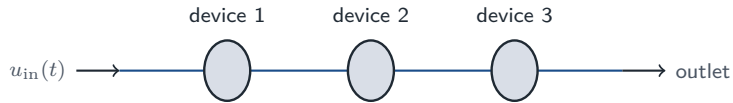
Governing dynamics, observations, reward

$$\partial_t u \;+\; u\,\partial_x u \;=\; \nu_0\,\partial_{xx} u \;+\; \underbrace{\alpha_{\text{eff}}(x,t)\,(u_\infty - u)}_{\text{recovery towards } u_\infty} \;-\; \underbrace{\sum_{j=1}^{N_a} \chi_j(t)\,n_j(x;\sigma)\,u_+(x,t)}_{\text{extraction at the } N_a \text{ devices}}$$



Governing dynamics, observations, reward

$$\partial_t u \;+\; u\,\partial_x u \;=\; \nu_0\,\partial_{xx} u \;+\; \underbrace{\alpha_{\text{eff}}(x,t)\,(u_\infty - u)}_{\text{recovery towards } u_\infty} \;-\; \underbrace{\sum_{j=1}^{N_a} \chi_j(t)\,n_j(x;\sigma)\,u_+(x,t)}_{\text{extraction at the } N_a \text{ devices}}$$



Observations (sparse, noisy):

$$\boldsymbol{o}_{u,k} = [u(0,t_k), u(x_1,t_k), \ldots],$$
$$\boldsymbol{o}_{\xi,k} = [\xi_{1,k}, \ldots]$$

Reward (extraction, balanced across devices):

$$r_k = \sum_{j=1}^{N_a} P_{j,k} - w_{\text{std}}\,\text{std}(\{P_{j,k}\})$$

Structural discrepancy: what the twin cannot know

Adaptive twin $\circ$ — each device acts on its own

$$\chi_{\circ,j} = \xi_j$$

a purely *additive* actuator model

Structural discrepancy: what the twin cannot know

Reality • — upstream loading degrades the devices behind it

$$\chi_{\bullet,j} = \xi_j \left[1 - \varepsilon \zeta_j\right]$$
$$\tau_\zeta \, \dot{\zeta}_j = -\zeta_j + \sum_{i<j} w_{ij} \, b_i$$

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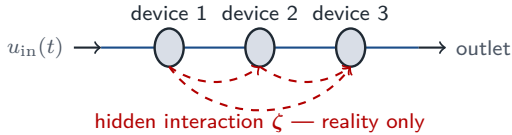

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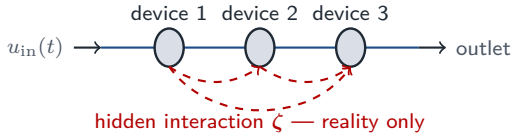

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Adaptive twin $\circ$ — each device acts on its own

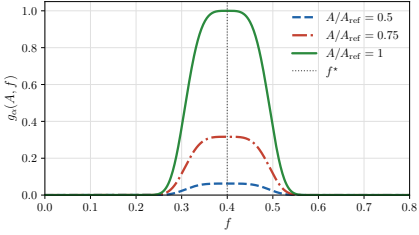
$$\chi_{\circ,j} = \xi_j$$

a purely *additive* actuator model

ζ is **never observed, never identified, never given to the twin** — so a *policy-dependent* model error survives even the best-fit twin.

The recovery coefficient depends on the actuator

$$\alpha_{\mathrm{eff}}(x,t) = \underbrace{\alpha_0}_{\text{static}} + \underbrace{q_{\mathrm{rec}}(x)\,\Delta\alpha\; g_\alpha\big(A_1(t), f_1(t)\big)\,\big[1+\mu_\alpha \sin(\phi(t)-\phi_\alpha)\big]}_{\text{dynamic — set by the upstream actuation } (A_1, f_1, \phi)}$$



$g_\alpha(A, f)$: a narrow band around $f^\star = 0.40$

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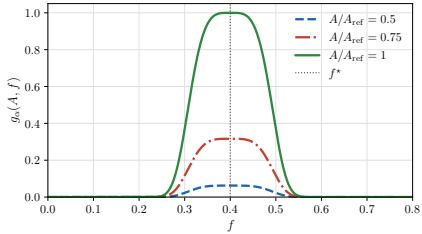
How fast the flow recovers depends on **what we are doing to it** — and with no modulation, $g_\alpha = 0$: quasi-steady again.



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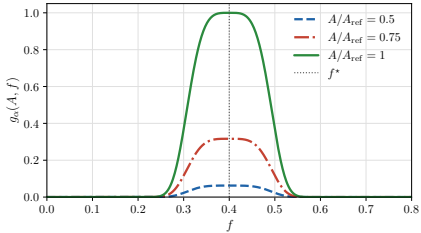
the six latent coordinates:

$$\boldsymbol{\theta}_p = [\,\alpha_0,\ \Delta\alpha,\ f^\star,\ s_f,\ f_\alpha,\ \sigma\,]^\top$$



$g_\alpha(A, f)$: a narrow band around $f^\star = 0.40$

The policy: a search in $\mathbb{R}^{13}$

Quasi-steady

$$\left[\pi_{\mathrm{qs}}\right]_j = \xi_{\max}\,\mathrm{sigm}\big(\beta_{j,0} + \beta_{j,1}\eta_k\big)$$

$$\eta_k = \tanh\big((z_k - u_{\mathrm{ref}})/s_z\big) \quad \text{with}$$
$$\tau_f \dot{z} = u_{\mathrm{in}}(t) - z$$

z is the causal low-pass filtered inlet — what the controller can actually see

a mean allocation per device, **conditioned on the inlet**

$\{\beta_{j,0}, \beta_{j,1}\}_{j=1,2,3}$ $\rightarrow$ **6 coordinates**

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Dynamic correction

$$d_{1,k} = h_D(A_{1,k}\sin\phi_k),$$
$$A_{1,k} = A_0 + k_A|z_k - u_{\mathrm{ref}}|, \quad f_{1,k} = f_0$$
$$d_{j,k} = A_j\sin(\phi_k - \psi_j), \quad j = 2,3$$

one upstream oscillator, two **phase-locked** followers

A_0, k_A, f_0 and $\{A_j, \psi_j\}_{j=2,3}$ $\rightarrow$ **7 coordinates**

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one upstream oscillator, two **phase-locked** followers

A_0, k_A, f_0 and $\{A_j, \psi_j\}_{j=2,3}$ $\rightarrow$ **7 coordinates**

$$\boldsymbol{\theta}_\pi = [\,\{\beta_{j,0}, \beta_{j,1}\}_{j=1,2,3},\ A_0,\ k_A,\ f_0,\ \{A_j, \psi_j\}_{j=2,3}\,]^\top \in \mathbb{R}^{13}$$

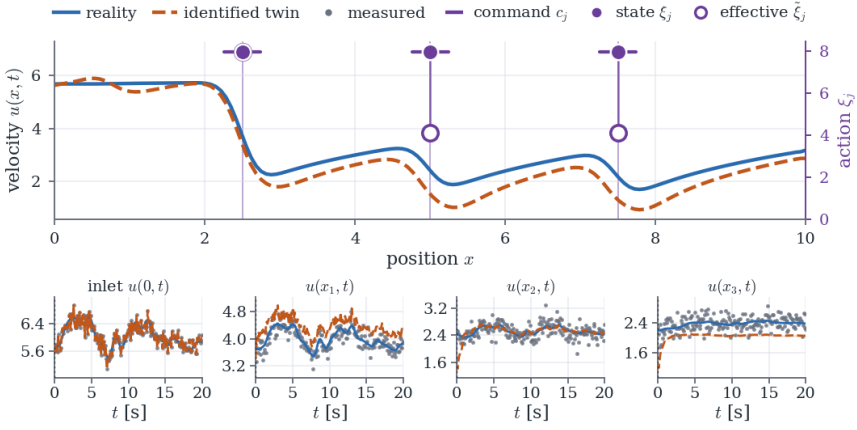
Section 5

Results

RESULTS

Baseline

Baseline: a simple constant actuation



No data, no twin — the only safe thing to do: a **constant, saturated command**. 50 such episodes (**40 train / 10 validation**) feed the replay identification:

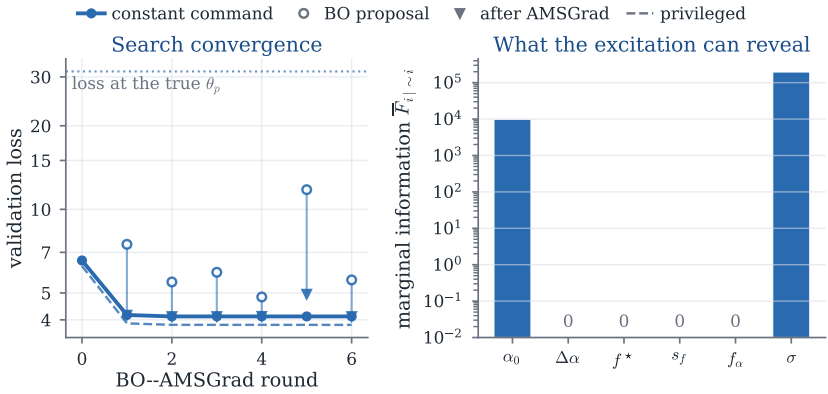
free forecast from the analyzed state, no correction over the scored window.

What that one policy is worth

Strategy	$\overline{P}_1$	$\overline{P}_2$	$\overline{P}_3$	$\sum_j \overline{P}_j$	std_j	$\widehat{\mathcal{R}}$	**episodes**
Constant, saturated	66.27	12.26	11.00	89.53	25.77	38.00	1

Brute force upstream **starves everything behind it** — and the balance penalty turns 89.5 of raw power into a return of $+\mathbf{38.0}$.

Convergence is not informativeness



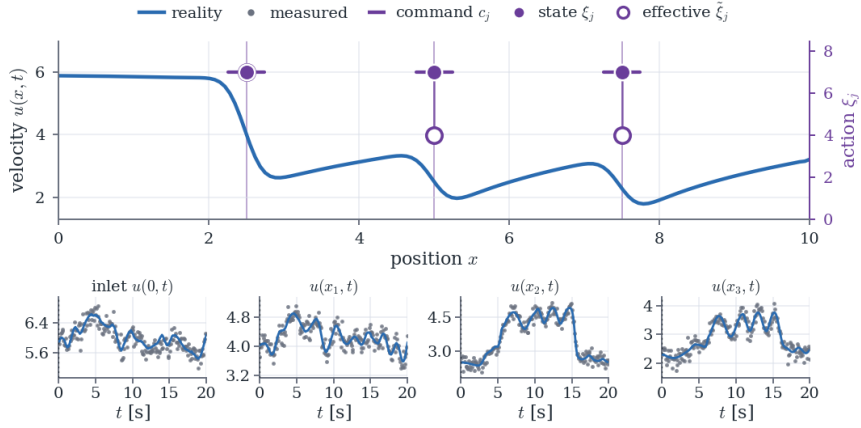
The fit converges in a handful of rounds — but four of the six coordinates carry **exactly zero** marginal information.

A converged fit is not the same as an identified model.

RESULTS

Model learning with exploration

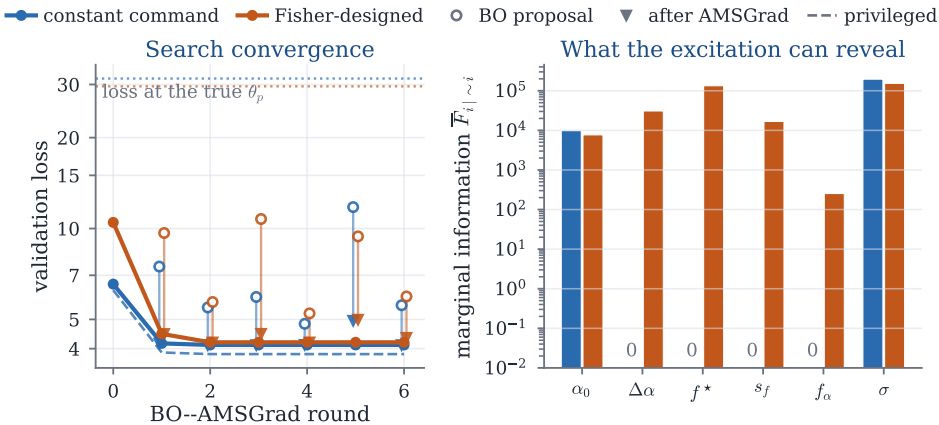
Now choose *what to do*, by information



$e_n = 1$: the command is chosen to maximize $\lambda_{\min}(\overline{\boldsymbol{F}})$. After 52 episodes: same blocks $[0.20, 0.32, 0.45, 0.58]$ Hz, the phase pattern the loop preferred.

The constant command became a modulated one.

Not a better a fit, but much more informative



$s_{\min}$: $\mathbf{0} \to \mathbf{15.90}$, and the validation loss does **not** move $(4.11 \to 4.20)$.

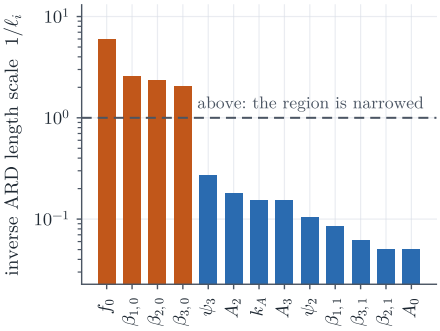
RESULTS

Model exploitation

An imperfect model: so what can we learn from it?

It tells us which knobs matter

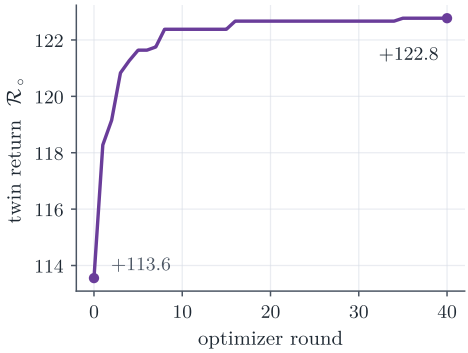
4 of 13 coordinates carry the search (spread 119×)



512 Sobol points on the twin's *own* returns — **no plant episode**

And it can propose a policy

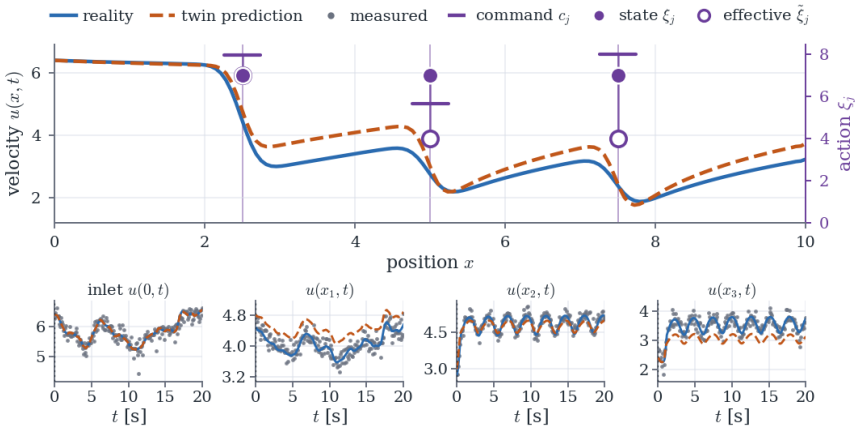
policy search inside the twin, zero plant episodes



optimised entirely inside the twin, on its own score

A search region and a starting point.

Not as good as expected



The twin's own prediction, drawn over the episode the plant actually delivered.

predicted $\mathcal{R}_\circ = +122.2$, **delivered** $\mathcal{R}_\bullet = +93.2$

RESULTS

Reinforcement twinning (and three ways an imperfect model helps and can be helped)

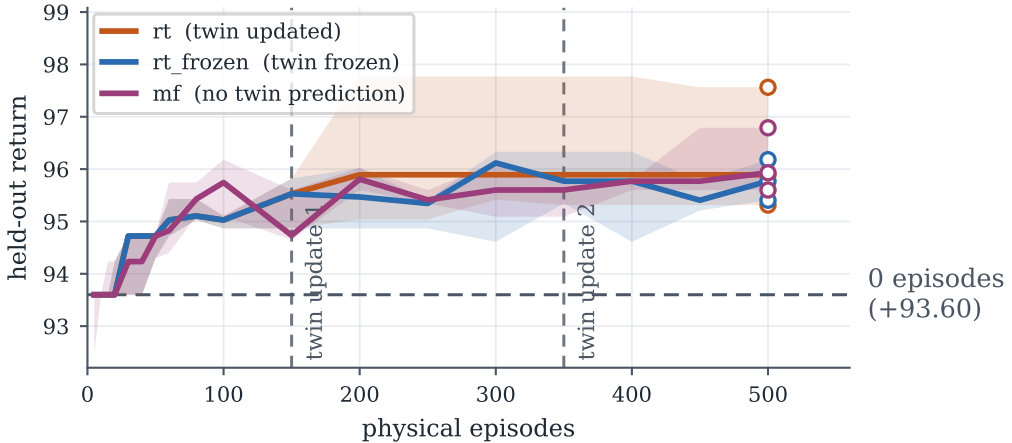
Different ways of model support

arm	trend basis $\phi_\circ$	twin	what it isolates
`mf`	$[1]$	—	the same search with $\gamma_{\circ,1} \equiv 0$
`rt_frozen`	$[1,\ \mathcal{R}_\circ]^\top$	frozen	is the twin's *prediction* worth a column?
`rt`	$[1,\ \mathcal{R}_\circ]^\top$	updated at 150, 350	the full reciprocal loop

Common: 500 episodes, the search centred on the twin's own best policy — every arm starts at $+93.60$; scored on eight held-out inflows.

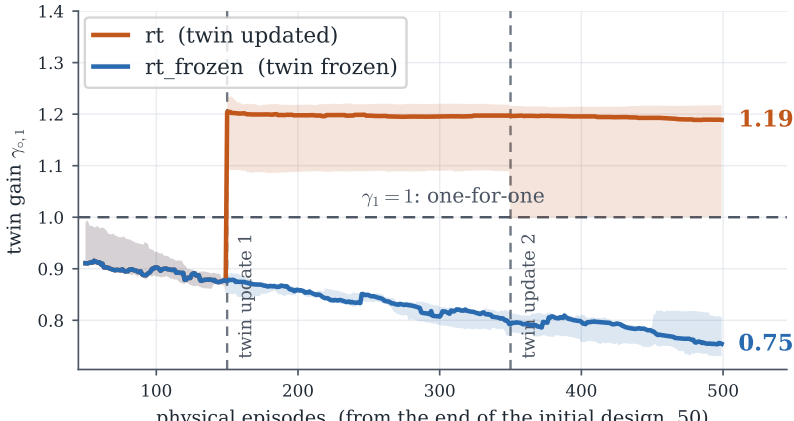
Only `rt` rebuilds the search at each twin update: new ARD metric, new centre, re-synthesised model-based policy.

Learning around the model



The model authority evolution

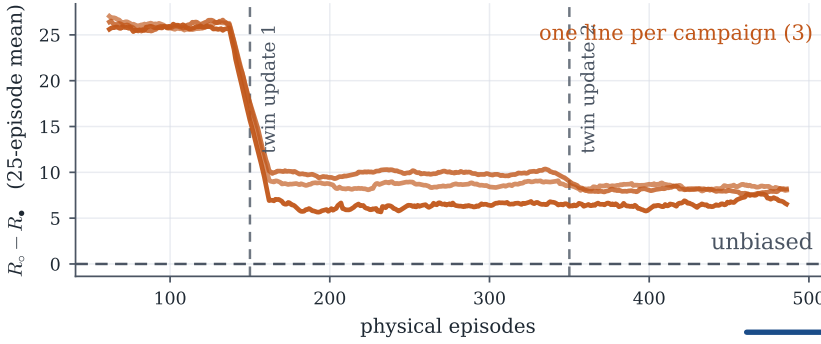
$$\mathcal{R}_\bullet(\boldsymbol{\theta}_\pi) = \underbrace{\gamma_{\circ,0} + \gamma_{\circ,1}\,\mathcal{R}_\circ(\boldsymbol{\theta}_\pi)}_{\text{calibrated twin trend}} + \delta_\circ(\boldsymbol{\theta}_\pi) + \varepsilon_\bullet$$



The model keeps improving where it matters

	α_0	$\Delta\alpha$	f^*	s_f	f_α	σ
truth	**0.5**	**6**	**0.4**	**0.1**	**0.08**	**0.2**
id., constant action	0.8	7.06	0.482	0.195	0.069	0.162
id., informative action	0.8	10	0.399	0.111	0.158	0.160
corrected (rt)	0.655	5.85	0.398	0.087	0.2	0.131

median, 6 campaigns	
replay loss $\mathcal{J}$	13.2 → 7.5 **−43%**
trajectory E_{dyn}	6.7 → 6.9 **flat**
return bias	+25.9 → +8.6 **−67%**



The scoreboard, complete

	policy	episodes	held-out	across campaigns
1	constant baseline	0	$+38.00$	—
2	identified from routine data	24	$+66.34$	—
3	**model-based, inside the twin**	**0**	$\mathbf{+93.60}$	—
4	`mf`	500	best $+96.79$	$+2.13 \pm 0.45$
5	`rt_frozen`	500	best $+96.70$	$+2.33 \pm 0.19$
6	`rt`	500	best $\mathbf{+97.56}$	$+2.55 \pm 0.32$

The last $+2.5$ is not the point: earning it corrected the model — driven by the one demand to predict outcomes well.

Section 6

Conclusions and Outlook

Conclusions

Even a wrong model is worth spending

- A twin with the *wrong structure* still delivers a policy worth $+\mathbf{93.6}$ with **zero** physical episodes.
- What it captures is **geometry**: four of thirteen knobs matter, and this survives even when the twin is rewritten.
- Sixty physical episodes on top of this prior outperform the twin's own policy in **8 seeds out of 8**.

And the plant pays it back

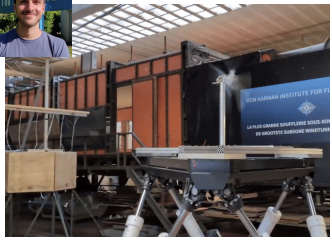
- Replay **corrects** the twin, pulling parameters away from their bounds and *toward the truth*, with essentially **one** update.
- The twin's authority is **inferred, never imposed**: $\gamma_{\circ,1}$ rises when the twin is corrected and decays when it becomes stale.
- This correction **triples** the twin's ability to rank policies.

A model good enough to be spent and a loop that makes it better.

S. Randino

Floating wind turbines

S. Randino

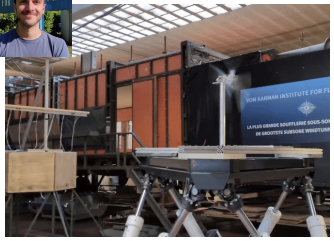


Floating wind turbines

L. Schena

Wind Farm Control

S. Randino



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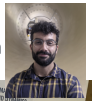
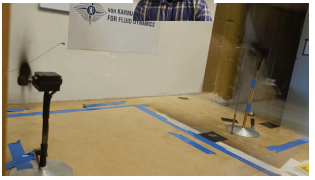
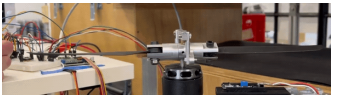




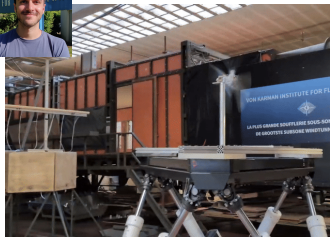
Wind Farm Control



Variable-pitch propellers

R. Poletti

S. Randino



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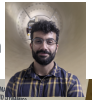

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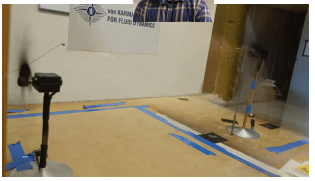

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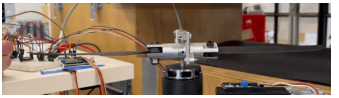

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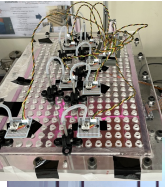


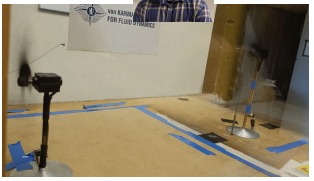
K.Bukowski, T. De Maria

Sloshing suppression

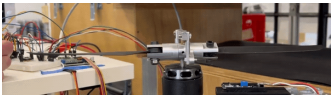
S. Randino

Floating wind turbines

L. Schena



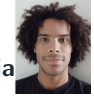
Wind Farm Control



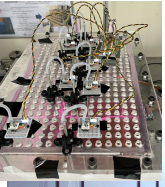
Variable-pitch propellers

R. Poletti

F. Monteiro, S. Ahizi



Cryogenic thermal storage



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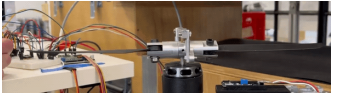
L. Schena

Wind Farm Control

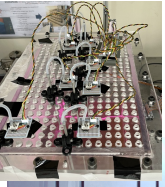


Variable-pitch propellers

R. Poletti

A. Grava and Y. Lecomte

Drone flight control



K.Bukowski, T. De Maria

Sloshing suppression

F. Monteiro, S. Ahizi

Cryogenic thermal storage

Thank you!

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ULB

Policy search between model-based and model-free extremes

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MODEL-BASED

plan inside a model

\+ sample efficient

– model bias

HYBRID / MODEL-GUIDED

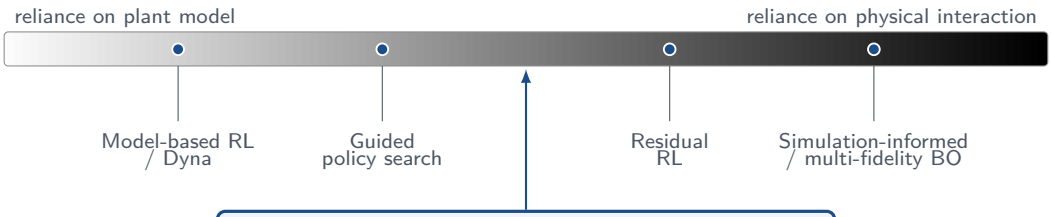
combine model knowledge with physical feedback

MODEL-FREE

learn from physical returns

robust to model error +

costly exploration –



Reinforcement Twinning

adaptive twin as a **calibrated prior**;

physical returns remain the reference

Filing the episode

The episode leaves behind one **factual record** $\mathcal{E}^{(n+1)}$ — the measured forcing, the actions actually applied, the observations, and the return achieved.

$$\mathcal{E}^{(n+1)} = \{\boldsymbol{z}_{0:N},\ \boldsymbol{a}_{\bullet,0:N},\ \boldsymbol{y}_{0:N},\ \mathcal{R}_{\bullet}\}$$

always $\rightarrow$ **replay buffer** $\mathcal{B}$

whatever the episode was deployed for

only if $e_n = 0 \rightarrow \mathcal{D}_\pi$

the *performance archive*: the pair $(\boldsymbol{\theta}_\pi, \mathcal{R}_\bullet)$, and the return surrogate is refitted

An identification episode is **not** a sample of the performance objective — so it never enters $\mathcal{D}_\pi$, but it still enriches the replay that will later update the twin.

Refreshing the guidance: what changes, what is kept

$u_{p,n}$ — identify

re-estimate $\hat{\boldsymbol{\theta}}_p$ from replay; recompute the analyzed states

$u_{\circ,\pi,n}$ — synthesize

advance the twin reference policy, warm-started from the previous solution

$u_{R,n}$ — recalibrate

rebuild the trend column with the new twin and refit $\widehat{\mathcal{R}}_\bullet$

The measured pairs $(\boldsymbol{\theta}_{\pi,i}, \mathcal{R}_{\bullet,i})$ are **never discarded**: only their twin predictions $\mathcal{R}_\circ^{(m)}(\boldsymbol{\theta}_{\pi,i})$ are recomputed as $\mathcal{R}_\circ^{(m+1)}$ before refitting.

So no two twin versions are ever mixed inside one calibration coefficient — and re-evaluating that column is **entirely virtual**: it costs no physical interaction.

A new proposal geometry applies only to *subsequent* candidates — nothing already measured is re-interpreted away.

Which policy acts, and what for?

Physical system $\bullet$ — the *deployed* policy is scheduled by e_n

$$\pi_{\bullet}^{(n+1)} = \begin{cases} \pi_{\bullet}(\cdot \mid \boldsymbol{\theta}_{\pi}^{(n+1)}), & e_n = 0 \quad \text{performance-oriented,} \\ \pi_{\iota}(\cdot \mid \boldsymbol{\theta}_{\iota}^{(n+1)}), & e_n = 1 \quad \text{information-oriented.} \end{cases}$$

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- $\pi_{\bullet}(\cdot \mid \boldsymbol{\theta}_{\pi})$ is the **persistent learning target**: the performance policy we are actually optimising.
- $\pi_{\iota}(\cdot \mid \boldsymbol{\theta}_{\iota})$ is a **transient excitation**, designed *virtually* and deployed for that one interaction only — it is not a second policy-learning problem.

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Adaptive twin $\circ$ — the action is not a control but a *declared source*

$$\boldsymbol{a}_{\circ,k} = \pi_\circ^{(n+1)} = \begin{cases} \boldsymbol{a}_{\bullet,k}, & \text{factual replay, synchronization, validation,} \\ \pi_\bullet(\boldsymbol{o}_{\circ,k} \mid \boldsymbol{\theta}_\pi), & \text{counterfactual evaluation of a candidate,} \\ \pi_\circ(\boldsymbol{o}_{\circ,k} \mid \boldsymbol{\theta}_{\circ,\pi}), & \text{model-based policy synthesis.} \end{cases}$$

Which policy acts, and what for?

Physical system $\bullet$ — the *deployed* policy is scheduled by e_n

$$\pi_\bullet^{(n+1)} = \begin{cases} \pi_\bullet(\cdot \mid \boldsymbol{\theta}_\pi^{(n+1)}), & e_n = 0 \quad \text{performance-oriented,} \\ \pi_\iota(\cdot \mid \boldsymbol{\theta}_\iota^{(n+1)}), & e_n = 1 \quad \text{information-oriented.} \end{cases}$$

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The first case **replays** the recorded action; the other two **close the loop on the twin's own observation** $\boldsymbol{o}_{\circ,k}$ — what makes a virtual experiment causal.

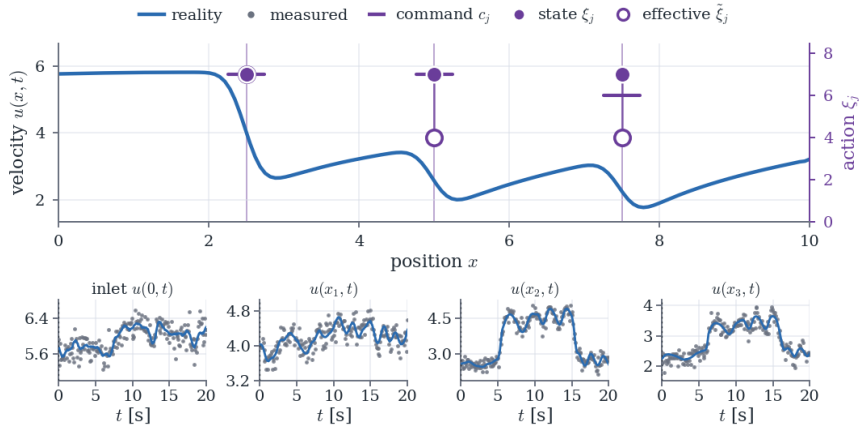
What those episodes can identify

		Baseline action	
Parameter	**planted**	**identified**	**equally-good range**
α_0	0.5000	0.8000	[0.800, 0.800]
$\Delta\alpha$	6.0000	7.0550	[4.373, 7.055]
$f^\star$	0.4000	0.4818	[0.317, 0.554]
s_f	0.1000	0.1949	[0.154, 0.195]
f_α	0.0800	0.0693	[0.069, 0.118]
σ	0.2000	0.1616	[0.161, 0.164]

Validation loss $\mathbf{4.11}$ — against $\mathbf{31.4}$ *at the planted truth*. And the state estimate is not the limitation: it is as good as a privileged one.

Four of the six could take almost any value and fit the data just as well.

The first informative design selected



$e_n = 1$: the command is now chosen to maximize $\lambda_{\min}(\overline{\boldsymbol{F}})$.

The *first* design selected — the devices oscillate about 7 instead of sitting at the corner.

The same six parameters, two excitations

		Baseline action		Explorative	
Parameter	**planted**	**identified**	**equally-good range**	**identified**	**equally-good range**
α_0	0.5000	0.8000	[0.800, 0.800]	0.8000	[0.800, 0.800]
$\Delta\alpha$	6.0000	7.0550	[4.373, 7.055]	10.0000	[8.532, 10.000]
$f^\star$	0.4000	0.4818	[0.317, 0.554]	0.3994	[0.395, 0.399]
s_f	0.1000	0.1949	[0.154, 0.195]	0.1109	[0.111, 0.118]
f_α	0.0800	0.0693	[0.069, 0.118]	0.1583	[0.105, 0.166]
σ	0.2000	0.1616	[0.161, 0.164]	0.1596	[0.159, 0.160]

The red is gone — every parameter is now pinned to a narrow interval.

Probing for information also controls better

Strategy	$\overline{P}_1$	$\overline{P}_2$	$\overline{P}_3$	$\sum_j \overline{P}_j$	std_j	$\widehat{\mathcal{R}}$	**episodes**
Constant, saturated	66.27	12.26	11.00	89.53	25.77	38.00	1
Fisher-designed excitation	69.39	26.15	16.72	112.26	22.93	66.34	52

Backing off saturation lifts downstream effectiveness $0.476 \to 0.528$, and the modulation lets the flow recover between devices — so the two rear devices more than double their output.

$+38 \to +66$**, from an experiment that was never about performance.**

What the model bought us, for zero physical episodes

Strategy	$\overline{P}_1$	$\overline{P}_2$	$\overline{P}_3$	$\sum_j \overline{P}_j$	std_j	$\widehat{\mathcal{R}}$	**episodes**
Constant, saturated	66.27	12.26	11.00	89.53	25.77	38.00	1
Fisher-designed excitation	69.39	26.15	16.72	112.26	22.93	66.34	52
Model-based policy search	72.92	31.27	29.29	133.48	20.12	93.22	0

$+\mathbf{93.6}$ held-out, **without a single new plant episode** — from $+38.0$ where we started.

The twin is optimistic about itself by 31%. It is still the best policy we have, and the cheapest.

Control and modelling as coupled learning processes

Controller: learn by intervening

- optimise physical performance
- decide which trajectories are explored
- generate data that may also be informative

Twin: learn by replaying

- predict the system evolution
- reconstruct the hidden dynamical state
- identify latent model parameters

Control and modelling as coupled learning processes

Controller: learn by intervening

- optimise physical performance
- decide which trajectories are explored
- generate data that may also be informative

Policy objective

$$\boldsymbol{\theta}_\pi^\star = \arg\max_{\boldsymbol{\theta}_\pi} \mathcal{R}_\bullet(\boldsymbol{\theta}_\pi \,;\, \boldsymbol{\theta}_p)$$

Twin: learn by replaying

- predict the system evolution
- reconstruct the hidden dynamical state
- identify latent model parameters

Identification objective

$$\boldsymbol{\theta}_p^\star = \arg\min_{\boldsymbol{\theta}_p} \mathcal{J}_\circ(\boldsymbol{\theta}_p \,;\, \boldsymbol{\theta}_\pi)$$

Step 2: deploy and record

1. Select the next interaction

performance $(e_n = 0)$ or information $(e_n = 1)$

2. Deploy and record

the only step that touches the physical system

3. Learn from the new experience

file the episode; synchronize; refit the return surrogate

4. Refresh model-dependent quantities

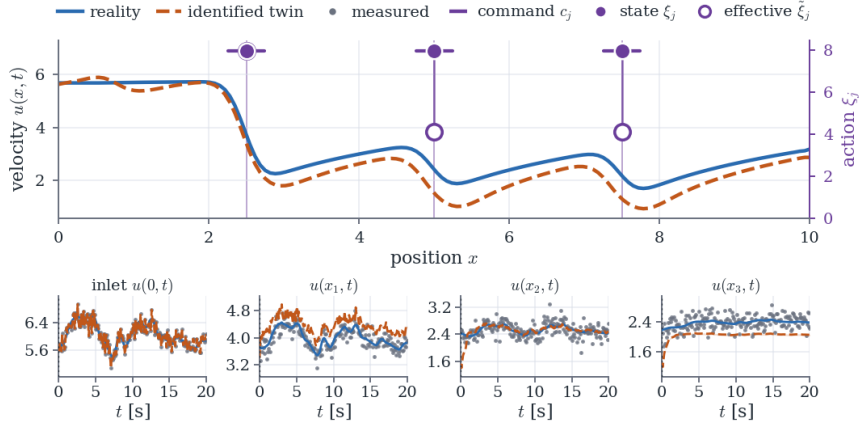
identify | re-synthesize | recalibrate

The only cost the loop pays — everything else is computation.

store $\mathcal{E}^{(n+1)} = \{\boldsymbol{z}_{0:N},\ \boldsymbol{a}_{\bullet,0:N},\ \boldsymbol{y}_{0:N},\ \mathcal{R}_\bullet\}$

policy-consistent: the replay will use the actions that were actually applied

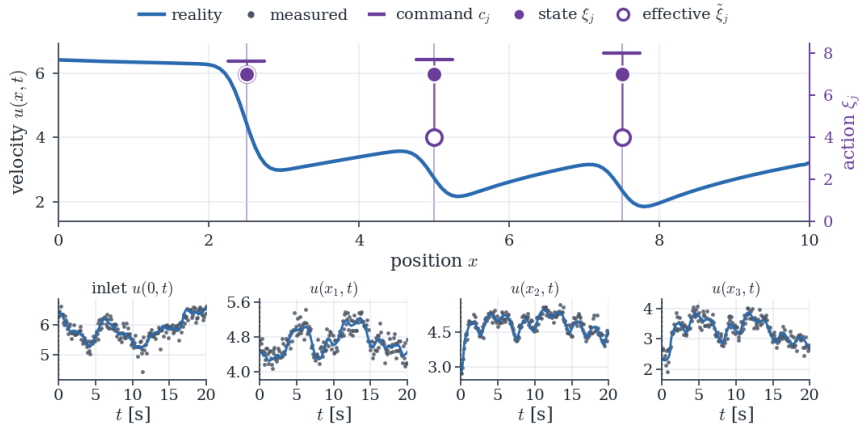
Identification from routine data



50 episodes of that same constant command, **40 train / 10 validation**.

Replay identification: free forecast from the analyzed state, no correction over the scored window.

The best policy we found, deployed



The best of all eighteen campaigns, on the plant. Same inflow and same axes as every earlier device E.